

2) Simplify the following expressions (leave your answers in index form)

(8 marks)

a) $12x^5 \div 3x^2$

$4x^3$

b) $(3x^5y)^4$

$81x^{20}y^4$

c) $\frac{3p^2q^5}{15p^6q^5}$

$\frac{1}{5p^4}$

d) $\frac{4x^2y \times (3xy^4)^2}{2x^4y^3 \times 3y^8}$

$= \frac{36x^4y^9}{6x^4y^{11}} = \frac{6}{y^2}$

3) Simplify the following expression

(10 Marks)

a) $\frac{4x}{7} + \frac{2x}{7}$

$\frac{6x}{7}$

b) $\frac{y}{5} - \frac{z}{4}$

$\frac{4y-5z}{20}$

c) $\frac{k}{9} \times \frac{3k}{2}$

$\frac{3k^2}{18} = \frac{k^2}{6}$

d) $\frac{x+3}{3} + \frac{2x+1}{5}$

$\frac{5(x+3) + 3(2x+1)}{15} = \frac{11x+18}{15}$

e) $\frac{2k+1}{2} \div \frac{k+7}{8}$

$\frac{2k+1}{2} \times \frac{8}{k+7} = \frac{8k+4}{k+7}$

f) $\frac{2t+7}{6} - \frac{3t+7}{2} \times \frac{t}{3}$

$\frac{2t+7}{6} - \frac{3t^2+7t}{6} = \frac{7-5t-3t^2}{6}$

4) Fully factorise the following expressions

(8 marks)

a) $3x^2 - 9x$

$3x(x-3)$

b) $x^2 + 13x + 30$

$(x+10)(x+3)$

c) $x^2 - 7x - 8$

$(x-8)(x+1)$

d) $y^2 - 36$

$(y-6)(y+6)$

e) $5x^2 - 50x + 120$

$5(x^2 - 10x + 24)$
 $5(x-6)(x-4)$

f) $98 - 2a^2$

$2(49 - a^2) = 2(7-a)(7+a)$

End of Test



Name: Sd

Score: /37
Total Time: 35 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used. 1 side A4 notes (handwritten) allowed.

1) Expand the following expressions

(11 marks)

a) $3(2x + 1)$

$6x + 3$

b) $-2x(x - 3)$

$-2x^2 + 6x$

c) $(x + 8)(x + 1)$

$x^2 + 9x + 8$

d) $(x + 3)(x - 2)$

$x^2 + x - 6$

e) $(a - 3)^2$

$a^2 - 6a + 9$

f) $(3p - 2)(5p + 11)$

$15p^2 - 10p + 11p - 22$
 $= 15p^2 + p - 22$

g) $(3x + 3)^2 - 2(x + 1)$

$9x^2 + 6x + 9 - 2x - 2$
 $= 9x^2 + 4x + 7$